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On a new Trichogrammatid (Hymenoptera: Chalcidoidea) *Uscana ranchiensis* sp.n. from Ranchi, Jharkhand

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Abstract : The present paper deals with a new Trichogrammatid belonging to the genus *Uscana* Girault. *U. ranchiensis* sp.n. differs from *U. alami* Yousuf & Shafee and U. mukerjii (Mani)

Key words: Trichogrammatidae, Hymenoptera, Chalcidoidea

INTRODUCTION

The family Trichogrammatidae belonging to the superfamily Chalcidoidea is a family of egg parasites of some important insect pests.

The genus *Uscana* was erected by Girault[1] in 1911 with its type species *Uscana semifumipennis* Girault[2] by monotypy. Another genus *Zaga* Girault which was also erected by him with the type species *Z. latipennis* by monotypy was synomymised by Yousuf & Shafee[3] with *Uscana* Girault in 1987.

MATERIAL AND METHODS

Collections were made by visiting important agricultural areas of Ranchi. Plant parts infested with insect eggs were cut into small pieces and put in rearing jars (4"×2"). The open end of the jars were covered with paper held with rubber bands. Attempts have also been made to rear the host up to adult stage for identification. A complete record of rearing was maintained indicating reference number, locality, date of collection, name of host plant and name of host insect. The collections were examined daily for the emerged parasites. The emerged parasites were preserved in 80% alcohol in glass vials. The preserved specimens were separated upto generic level under the binocular with the help of fine needles. Collections were also made by sweeping the vegetation in the field.

The permanent slides were prepared to enable detail study of important structures of the parasites. The normal process of dehydration was adopted and clearing was done in clove oil. The specimens were dissected in clove medium. The dissected parts were placed on a micro-slide in a drop of canada balsam and oriented to the required position. The slides were allowed to dry for some time. This was followed by adding required quantity of canada balsam to the slides and cover slips were placed. The slides were put in thermostat for five to six days to make them completely dry.

Drawings of important structures were made with the help of camera lucida. Measurements of whole insects were made with the help of an ocular micrometer.

Genus *Uscana* Girault, 1911

Uscana Girault[4], 1911d : 22.

Type-species: *Uscana semifumipennis* Girault, by monotypy.

Zaga Girault[4], 1911d : 30.

Type-species ; *Zaga latipennis* Girault, by monotypy.

Bruchoctonus Greese[5], 1923 : 117. Syn. quoted by Peck[6], 1963 : 82.

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Type-species : *Bruchoctonus senex* Greese[5], by monotypy.

Diagnosis : Antennae without funicle, club 4-segmented; fore wings with stigmal vein well developed, about as long as premarginal vein; discal setae almost arranged in rows, vein track RS_1 present; male genitalia (Viggiani[7], 1971; fig. VIII : 7) simple, tubular without digiti and claspers.

Comments : The genera *Uscana* Girault[4] and *Zaga* Girault are closely related on the basis of number of characters viz., 4-segmented antennal club, arrangement of discal setae in rows and presence of vein track RS_r Therefore, *Zaga* Girault[4] is accepted here as synonym of *Uscana* Girault[4].

The genus *Uscana* Girault[4] is known to contain 16 species including one new species. 2 species have been reported from India and a key for their separation is given below :

Key to Indian species of *Uscana* Girault[4], based on females

1. Fore wings slightly more than two times as long as wide, marginal fringe one-fourth the wing width; antennal club three times as long as wide, with long setae
1. *alami* Yousuf & Shafee[3]

— Fore wings (fig. 1A) two and a half times as long as wide, marginal fringe one-third the wing width; antennal club (fig. 1B) slightly more than two times as long as wide, with short setae
2. *ranchiensis* sp. n.

DESCRIPTION

1. *Uscana alami* Yousuf & Shafee[3]
Chaetostricha mukerjii Mani[8]*, 1935b : 337.*
Uscana mukerjii (Mani); Doutt & Viggiani[9], 1968 : 554.
Hosts: Bruchus quadrimaculatus Fabricius.
Callosobruchus analis (Fabricius)
Callosobruchus chinensis (Linnaeus)
Callosobruchus maculatus (Fabricius)
Distribution : INDIA : Calcutta, Chandigarh.

2. *Uscana ranchiensis* sp. n, (Fig. 1A,B)
Female.

Head yellowish brown, slightly longer than wide in facial view; orange; ocelli red. Antennae (fig. 1A) yellow; scape cylindrical, three times as long as wide; pedicel slightly longer than wide; single ring segment present; funicle absent; club 4-segmented, about two and a half times as long as wide, apical segment longest, segments with long setae.

Thorax yellow. Fore wings (fig. 1B) hyaline except an infuscated patch beneath premarginal vein, about two and a half times as long as wide; discal setae almost arranged in rows; vein track RSj present with numerous setae, reaching beneath the premarginal vein; costal cell narrow; post marginal vein rudimentary; stigmal vein well developed, about as long as premarginal vein. Legs yellow except femora of mid and hind legs dark brown.

Abdomen yellow, slightly longer than thorax; ovipositor hidden, arising from base of abdominal venter.

Body length ; 0.56 mm.

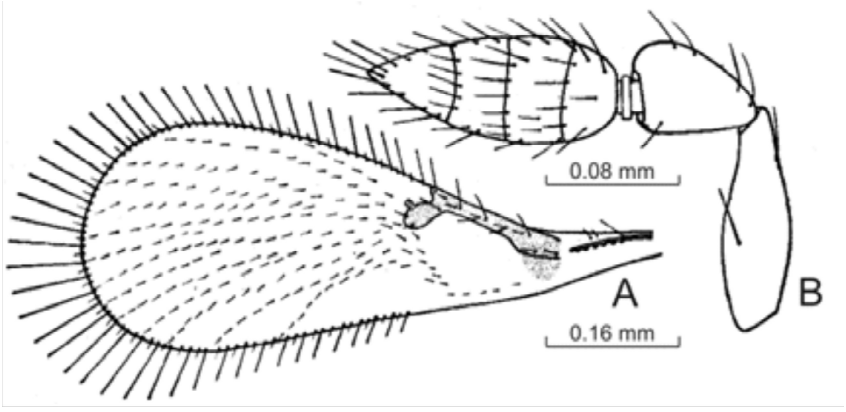


Fig. 1 A-B, *Uscana ranchiensis* sp. n. : A, fore wing ♀ ; B, antenna, ♀

Holotype♀. India : Jharkhand, Ranchi, 04.11.2013, by rearing as well as sweeping, Gulnaz Shamim.

DISCUSSION

Uscana ranchiensis sp. n. is closely related to ***U. alami*** Yousuf & Shafee[3], but can be distinguished by having fore wings two and a half times as long as wide, marginal fringe one-third the wing width; antennae with club slightly more than two times as long as wide, with short setae.

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REFERENCES

1. **Girault, A. A. 1911a**. Synonymic and descriptive notes on the chalcidoid family Trichogrammatidae with descriptions of new species. *Trans. Am. Entomol. Soc*. **37**: 43-83.
2. **Girault, A.A. 1911c**. On the identity of the most common species of the family Trichogrammatidae (Hymenoptera). Bull. Wisconsin. *Nat. hist. Soc*. **9** : 135-165.
3. **Yusuf, M. & Shafee, S. A. 1985c.** Descriptions of two new species of Trichogrammatidae (Hymenoptera : Chalcidoidea) from India. *Bull. Soc. Entomol. Suisse*, Switzerland, **58** : 303-305.
4. **Girault, A. A. 1911d**. Descriptions of nine new genera of the chalcidoid family Trichogrammatidae. *Trans. Am. Entomol. Soc*. 37 : 1-42.
5. **Greese, 1923.** Bull. Saratov Seed select. Admin. Sugar Trust. Kiev. 7 : 117. Haliday, A.H. 1833*. *Entomol. Mag.* 1 : 340.
6. **Peck, O. 1963**. A catalogue of the Nearctic Chalcidoidea (Insecta : Hymenoptera). *Can. Entomol.* **30** : 1092 pp.
7. **Viggiani, G. 1971a**. A new species of Chaetostricha Walk. from Africa, XXVI. Researches on the hymenoptera Chalcidoidea (Trichogrammatidae). *J. Entomol. Soc*. South Africa **34** : 33-35.
8. **Mani, M. S. 1935b**. First record of Trichogrammatid chalcid genus Chaetostricha Walker from India with description of new species. *Rec. Indian Mus*. **37** : 337-338.
9. **Doutt, R. L. & Viggiani, G. 1968**. The classification of the Trichogrammatidae (Hymenoptera : Chalcidoidea). *Proc. Calif. Acad. Sci*. **35** : 477-586.

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